

Vilal Ali

Hyderabad, India

Mobile: +91 9984736691

Email: vilal.ali@research.iiit.ac.in

Alt Email: belaal.ali@gmail.com

LinkedIn: linkedin.com/in/vilal-ali

GitHub: github.com/vilalali

Portfolio: vilalali.github.io/vilal-ali

Personal Profile

Lead Software Engineer and Solution Architect with 10+ years of experience designing scalable enterprise systems, AI platforms, and full-stack applications. Proven track record of architecting microservices-based solutions, leading cross-functional engineering teams, and delivering production systems from concept to deployment. Expertise in cloud architectures, distributed systems, and AI-enabled platforms across health-care, agriculture, and national data initiatives. Published the paper “AH-Translit: A Multi-Domain Dataset and Benchmark for Arabic-to-Hindi Transliteration” at the NeurIPS 2025 Affinity Workshop.

Experience

Feb 2023 – Present: Sr. Software Engineer — Tech Lead (AI Platform & Full Stack)

IHub Data, IIIT Hyderabad

- Architected scalable AI microservice platforms supporting model hosting and dataset management.
- Built and deployed LLM-powered chatbots and AI assistants integrated into production web platforms.
- Developed a histopathology large-image viewer and machine learning challenge platforms.
- Led a team of 12 members, conducting code reviews, system design decisions, and technical mentoring.
- Collaborated with cross-functional teams to deliver end-to-end production deployment of systems.

Apr 2019 – Feb 2023: Software Engineer (NLP & AI Systems)

eBhasha Setu, CIE, IIIT Hyderabad

- Developed multilingual NLP and transliteration systems and deployed on production.
- Built ML pipelines for transliteration, OCR processing, ASR transcription, and translation workflows.
- Developed Videozaar an AI-based audio/video transcription and subtitle synchronization platform.
- Winner of Bhashini Challenge, 2024 Govt. documentation digitization in Indian regional languages.

Sept 2018 – Apr 2019: Research Associate (AI Applications)

CVEST Research Center, IIIT Hyderabad

- Built research-grade AI web platforms using Flask and full-stack architectures.
- Developed desktop research tools using ElectronJS for applied AI workflows.

July 2014 – Sept 2018: Computer Science Lecturer

Unity Pvt. ITI, Lucknow, UP

- Taught core computer science subjects including programming, algorithms, and software fundamentals.
- Developed curriculum, practical labs, and project-based learning modules.
- Mentored students on technical projects and problem-solving approaches.
- Designed assessments and evaluated academic performance.

Research & Projects

India Data Org — National AI Data Platform

- Led conceptual design of a national-scale data platform for secure dataset hosting, an model hosting.
- Defined data structures, and integration mechanisms for heterogeneous datasets across domains.
- Established scalable microservice architecture enabling data ingestion, validation, and controlled access.
- Integrated AI-powered tools into production systems to support research and deployment initiatives.
- Project Tech Stack: Node.js, React.js, Microservices, MySQL, and AWS architecture.

Crop Darpan — AI based Disease Diagnostic System (Patent Granted)

- Designed a visual symptom-driven agricultural decision support system that mimics expert diagnostic reasoning to provide real-time crop disease identification and advisory via smartphone.
- Established methodologies for collecting, validating agricultural data from domain experts.

- Extended for explainable image-based diagnosis using MobileViT models optimized for Qualcomm QIDK.
- Project Tech Stack: Python, Flutter, Machine Learning, MobileViT, MySQL, REST APIs, and Edge AI.
- Project Repo: github.com/cropdarpan

CXR2ScribeAI — Automated Chest X-ray Report Generation

- Conducted comparative study of baseline, two-stage pipeline, and end-to-end multimodal architectures for automated radiology report generation.
- Fine-tuned and evaluated models on MIMIC-CXR dataset using 13 quantitative & clinical concept metrics.
- Designed evaluation workflows for clinically grounded report generation and model comparison.
- Project Tech Stack: Python, PyTorch, Transformers, Medical AI, MIMIC-CXR, and Deep Learning.
- Project Repo: github.com/vilalali/CXR2ScribeAI.git

MediLingua — Multilingual Safe Medical QA System (RAG + Safety Guardrails)

- Built a multilingual medical Question Answering system using RAG over curated medical knowledge sources.
- Designed safety guardrail pipeline medical safety classifiers queries.
- Cross-lingual QA workflow using embedding retrieval + LLM generation + translation-in/out architecture.
- Secure, fact-grounded medical response generation to reduce hallucinations in healthcare AI settings.
- Project Tech Stack: Python, RAG, LLMs, FAISS, NLP, Translation Models, and Cloud-Native Services.
- Project Repo: github.com/vilalali/mediLingua

Histopathology Viewer for Large File Visualization

- Developed a Machine Learning Challenge platform to track and rank participant performance.
- Integrated the Histopathology Viewer as a microservice into the core Data Foundation platform.
- Enhanced the platform's capability to handle large-scale medical image with optimized performance.
- Project Tech Stack: Node.js, React.js, Microservices, OpenSeadragon, MySQL, Docker, and AWS.

Audio/Video Transcription Tool: Videozaar

- Built a web application using ASR engine to transcript the task.
- Write a FASTAPI to convert video into the audio format in the required format.
- Integrate ASR APIs with the Videozaar.
- Project Tech Stack: Python, FastAPI, ASR, FFmpeg, NLP, React.js, and AWS.

Academic Publications

AH-Translit: A Multi-Domain Dataset and Benchmark for Arabic-to-Hindi Transliteration

Vilal Ali et al., NeurIPS 2025 Affinity Workshop, Paper Link

Engineering & Research Expertise

Core Competencies: System Architecture, Microservices, Cloud Platforms, Distributed Systems, AI Platform Design, Full-Stack Development, Technical Leadership, Large-Scale Data Systems.

Tech Stack: Python, PyTorch, Flask, FastAPIs, Node.js, React, REST APIs, Redis, MongoDB, MySQL, Docker, CI/CD, AWS.

AI/ML Expertise: Data Platform Design, Data Pipelines, AI/ML Systems for Real-World, Agricultural AI, Healthcare AI, Multimodal & NLP Systems, Model Evaluation, Microservices, Cross-Functional Team Collaboration.

Education

2024 – 2026: MS By Research

International Institute of Information Technology Hyderabad (IIIT-H)

Focus: NLP Systems, Multilingual AI, AI in Healthcare, AI in Agriculture, Applied ML, Data Science & Analytics.

2020, 2023: PGSSP in System Design and Development, Software Engineering

International Institute of Information Technology Hyderabad (IIIT-H)

Focus: System Design and Development, Software Engineering

2012 – 2015: Master of Computer Applications (MCA)

Institute of Engineering & Technology, India

Focus: Software Engineering, Algorithms, Distributed Systems

2007 – 2010: B.Sc. in Computer Science

Focus: Mathematics, Computer Science, Physics

Lucknow University, India